

In **R programming**, an **object** is anything that stores data.

Everything in R is treated as an object — numbers, text, lists, functions, datasets, even models

What is an Object in R?

An **object** is a named storage space that holds a value.

👉 Think of it like a **container with a name**.

Example:

```
R
x <- 10
```

Here:

- `x` is the object name
- `10` is the value stored in it
- `<=` is the assignment operator

Now `x` contains 10.

How to Create an Object

You can create objects using:

1 `<=` (Most commonly used)

```
R
a <- 5
```

2 `=` (Also works)

```
R
b = 20
```

3 `assign()` function

```
R
assign("c", 30)
```

Types of Objects in R

R has different types of objects.

1 Numeric Object

```
R
```

```
R  
  
name <- "Ajay"
```

3 Logical Object

```
R  
  
flag <- TRUE
```

4 Vector Object

```
R  
  
v <- c(1, 2, 3, 4)
```

5 Matrix Object

```
R  
  
m <- matrix(c(1,2,3,4), nrow=2)
```

6 List Object

```
R  
  
mylist <- list(1, "Ajay", TRUE)
```

7 Data Frame Object

```
R  
  
df <- data.frame(name=c("A", "B"), age=c(20, 22))
```

Checking Object Type

Use these functions:

```
R  
  
class(x)  
typeof(x)  
str(x)
```

Example:

```
R  
  
x <- 10  
class(x)
```

Output:

```
arduino
```

Rules for Naming Objects

- ✓ Can contain letters, numbers, dot (.) and underscore (_)
- ✓ Must start with a letter or dot
- ✗ Cannot start with a number
- ✗ Cannot use reserved words like `if`, `for`, `while`

Valid:

```
R
my_data
age1
.total
```

Invalid:

```
R
1age
for
```

Removing an Object

```
R
rm(x)
```

Remove multiple objects:

```
R
rm(x, y, z)
```

Remove all objects:

```
R
rm(list = ls())
```

Viewing All Objects

```
R
ls()
```

Important Concept: Everything is an Object

In R:

- Functions are objects
- Models are objects

mean

mean itself is an object (a function).

Simple Example Program

```
R

# Creating objects
name <- "Ajay"
age <- 22
marks <- c(80, 85, 90)

# Calculating average
average <- mean(marks)

# Display
print(average)
```

In Very Simple Words

An **object** in R is:

A named container that stores data or values.

Without objects, you cannot store or work with data in R.

If you want, I can also explain:

- Object vs Variable in R
- Data types in detail
- Attributes of objects
- OOP in R (S3, S4, Reference classes)

explain about type

In **R programming**, a **vector** is the most basic data structure.

A vector stores **multiple values of the same type** in a single object.

 Important rule:

All elements in a vector must be of the **same data type**.

Types of Vectors in R

There are mainly **6 types of vectors** in R.

```
x <- c(TRUE, FALSE, TRUE)
```

Check type:

```
R
```

```
typeof(x)
```

Output:

```
arduino
```

```
"logical"
```

Used in conditions and comparisons.

2 Integer Vector

Stores whole numbers.

```
R
```

```
x <- c(1L, 2L, 3L)
```

⚠ In R, you must use `L` to specify integer.

Check:

```
R
```

```
typeof(x)
```

Output:

```
arduino
```

```
"integer"
```

3 Numeric (Double) Vector

Stores decimal or real numbers.

```
R
```

```
x <- c(10, 20.5, 30)
```

Default numeric type in R is **double**.

Check:

```
R
```

```
typeof(x)
```

4 Character Vector

Stores text values.

```
R
names <- c("Ajay", "Ravi", "Kiran")
```

Check:

```
R
typeof(names)
```

Output:

```
arduino
"character"
```

5 Complex Vector

Stores complex numbers (real + imaginary).

```
R
x <- c(2+3i, 4+5i)
```

Check:

```
R
typeof(x)
```

Output:

```
arduino
"complex"
```

6 Raw Vector

Stores raw bytes (rarely used in basic programming).

```
R
x <- charToRaw("A")
```

Check:

```
R
typeof(x)
```

"raw"

Important Concept: Type Conversion (Coercion)

If you mix different types in a vector, R converts everything to one common type.

Example:

R

```
x <- c(1, "Ajay", TRUE)
```

Output:

arduino

```
"1" "Ajay" "TRUE"
```

👉 R converts everything to **character**.

Coercion order (low to high):

sql

```
logical → integer → double → character
```

Summary Table

Type	Example	Description
Logical	TRUE	TRUE/FALSE
Integer	5L	Whole numbers
Numeric	5.5	Decimal numbers
Character	"A"	Text
Complex	2+3i	Complex numbers
Raw	raw()	Bytes

In Very Simple Words

Vector types in R depend on **what kind of data** they store:

- Numbers → Numeric / Integer
- Text → Character
- TRUE/FALSE → Logical
- Real + imaginary → Complex
- Binary → Raw

- Vector operations
- Difference between vector and list

Just tell me 👍



